Equity-Aware Ambulance Outpost Siting in Sarıyer

INDR 486/586 Term Project

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Abstract

This project develops a district-scale ambulance outpost siting model for Sarıyer, Istanbul. The study uses selected neighborhoods as demand points, population as demand weight, public buildings as candidate ambulance outposts, and driving distance as the accessibility measure. The model minimizes population-weighted average distance while evaluating equity through maximum distance, 90th percentile distance, weighted coefficient of variation, and weighted Gini coefficient. The computational results indicate that opening four outposts provides a strong balance between efficiency, equity, and implementation cost.

1 Introduction

Emergency medical service planning requires decisions about where ambulances or ambulance outposts should be located so that response access is both efficient and equitable. In a district such as Sarıyer, spatial heterogeneity is important because neighborhoods differ in population, geography, and road accessibility.

This project studies a simplified but reproducible ambulance outpost siting problem for Sarıyer. The methodology follows the general facility-location logic used by Görmez, Köksalan, and Salman [1], who studied disaster response facility location in Istanbul using road-network distances and bi-criteria optimization.

2 Data

2.1 Demand Points

The demand points are ten selected neighborhoods in Sarıyer. Population is used as the demand weight. The neighborhood population values were obtained from publicly available Sarıyer neighborhood population data derived from the Turkish Address Based Population Registration System [2, 6].

2.2 Candidate Facilities

Seven candidate ambulance outpost locations were selected from public or quasi-public facilities in Sarıyer, including hospitals, a polyclinic, a municipality building, a stadium, a fire station, and a mukhtarlık. These locations are not claimed to be official ambulance stations; they are candidate outpost sites used for the optimization exercise.

2.3 Distance Matrix

The model uses driving distance between each demand point and each candidate facility. The reproducible project includes a seed distance matrix in kilometers. The intended routing workflow is based on OpenStreetMap road data and OSRM routing services [4, 5]. Geocoding should respect Nominatim usage policies, including request-rate limits and caching [3].

3 Mathematical Model

Let I be the set of demand points and J the set of candidate facilities.

3.1 Parameters

- w_i: population of demand point $i \in I$
- d_{ij}: driving distance from demand point i to facility j
- P: maximum number of facilities allowed to open

3.2 Decision Variables

$$y_j = \begin{cases} 1, & \text{if facility } j \text{ is opened} \\ 0, & \text{otherwise} \end{cases}$$

$$x_{ij} = \begin{cases} 1, & \text{if demand point } i \text{ is assigned to facility } j \\ 0, & \text{otherwise} \end{cases}$$

3.3 Objective Function

The main objective minimizes the population-weighted average distance:

$$\min \frac{\sum_{i \in I} \sum_{j \in J} w_i d_{ij} x_{ij}}{\sum_{i \in I} w_i}$$

3.4 Constraints

Each demand point must be assigned to exactly one opened facility:

$$\sum_{j \in J} x_{ij} = 1 \quad \forall i \in I$$

A demand point cannot be assigned to a closed facility:

$$x_{ij} \leq y_j \quad \forall i \in I, j \in J$$

The number of opened facilities is limited by P:

$$\sum_{j \in J} y_j \leq P$$

The variables are binary:

$$x_{ij}, y_j \in \{0, 1\}$$

4 Equity Metrics

In addition to average weighted distance, the project evaluates equity using four indicators:

- Maximum assigned distance
- Weighted 90th percentile distance
- Weighted coefficient of variation
- Weighted Gini coefficient

These metrics capture whether some neighborhoods remain poorly served even when the average distance is low.

5 Computational Implementation

The model was implemented in Python. The project uses CSV input files for demand points, candidate facilities, and the distance matrix. The solution procedure evaluates all possible facility subsets for each value of $P = 1, \ldots, 7$. Because the number of candidate facilities is small, enumeration is exact and computationally simple.

6 Results

Table 1 reports the efficient frontier from the seed distance matrix.

Table 1: Efficient frontier and equity metrics

MaxFac	Opened Facilities	Avg. km	Max km	P90 km	CV	Gini
1	F4	5.3273	14.4000	14.4000	0.7421	0.3717
2	F4;F7	3.2804	5.0000	4.8000	0.3662	0.2039
3	F2;F3;F7	2.4109	5.0000	4.6000	0.6388	0.3593
4	F2;F3;F6;F7	2.0585	3.6000	3.4000	0.5798	0.3237
5	F2;F3;F4;F6;F7	1.7397	3.4000	2.8000	0.5815	0.3275

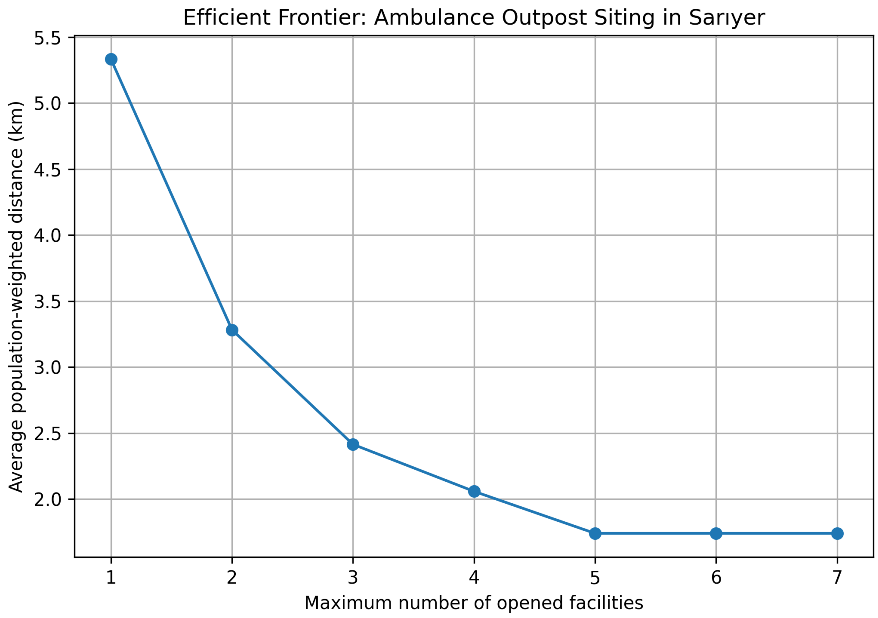

Figure 1: Efficient frontier for ambulance outpost siting in Sarıyer

6.1 Sensitivity Analysis

To evaluate the robustness of the recommended solution, two sensitivity analyses were conducted. First, the maximum number of opened facilities was varied from 1 to 7. Second, different service-distance thresholds were tested to measure the percentage of population covered within each threshold.

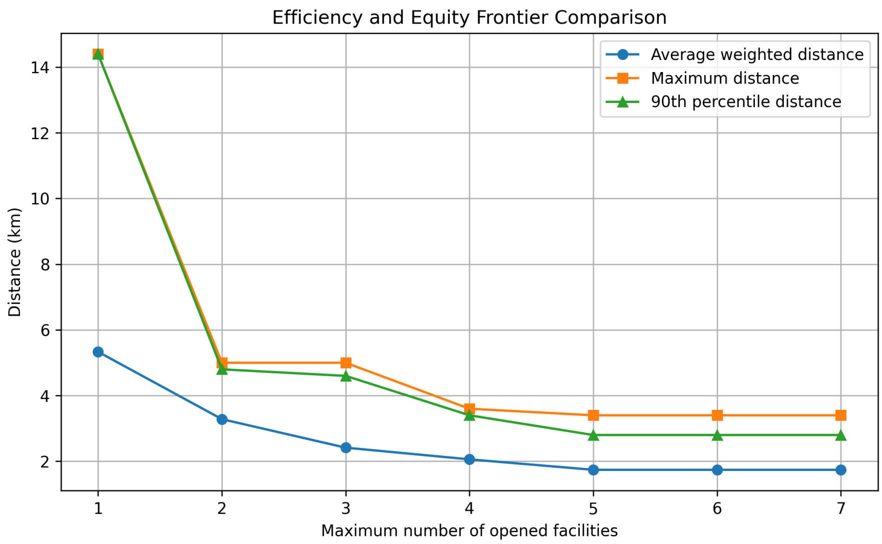

Figure 2: Comparison of average, maximum, and 90th percentile service distances

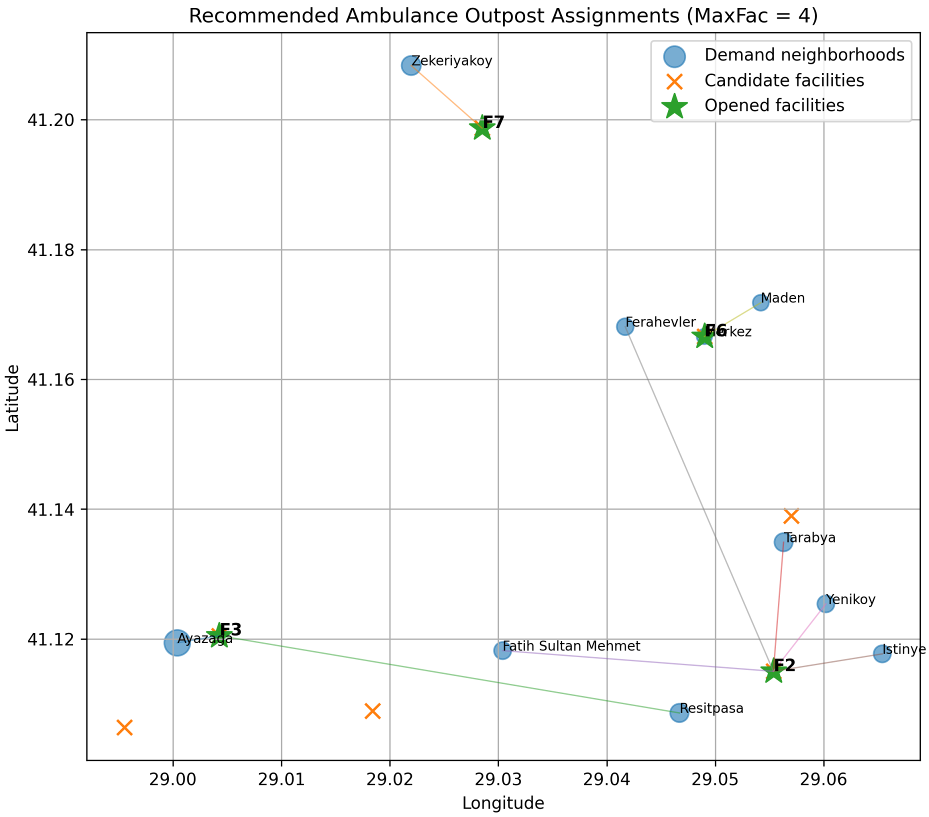

Figure 3: Recommended assignment map for MaxFac = 4

7 Recommended Solution

The recommended solution is to open four facilities:

$$\{F2, F3, F6, F7\}$$

This solution provides a strong practical compromise. Compared with three facilities, opening the fourth facility reduces the maximum assigned distance from approximately 5.0 km to 3.6 km. The fifth facility further improves the average distance, but the improvement is smaller relative to the additional infrastructure requirement.

Table 2: Recommended assignments for MaxFac = 4

Demand Neighborhood	Assigned Facility	Distance (km)
Ayazaga	F3	0.4
Zekeriyakoy	F7	1.8
Resitpasa	F3	3.4
Tarabya	F2	3.4
Fatih Sultan Mehmet	F2	3.6
Istinye	F2	0.4
Yenikoy	F2	2.0
Ferahevler	F2	2.8
Maden	F6	2.4
Merkez	F6	2.3

8 Limitations

This project is a district-scale planning model, not an official EMS deployment plan. The model uses population as a proxy for EMS demand, whereas real ambulance demand depends on call histories, age distribution, health conditions, time of day, and traffic. The seed distance matrix should ideally be replaced by live OSRM or openrouteservice driving distances before final submission.

9 Conclusion

The analysis shows that an equity-aware ambulance outpost siting model can be implemented for Sarıyer using public data and open-source tools. The recommended four-facility solution improves both efficiency and equity, while avoiding unnecessary facility expansion. This makes it a suitable practical recommendation for the course project.

References

[1] N. Görmez, M. Köksalan, and F. S. Salman. Locating disaster response facilities in istanbul. *Journal of the Operational Research Society*, 62(7):1239–1252, 2011.

[2] Nufusu.com. Sarıyer mahalleleri nüfusu, 2026. Accessed for neighborhood-level Sarıyer population values.

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[6] Turkish Statistical Institute. Address based population registration system, 2026. Official population registration framework.